

Census Income Classification and Customer Segmentation for Marketing

Executive Summary

This project addressed two marketing questions for the client: **who should be targeted** and **how those people should be approached**.

To answer the first question, I built an income classification model to identify individuals likely to earn **more than \$50K**. Two models were tested: a Logistic Regression benchmark and a CatBoost model. **CatBoost is the recommended production model** because it achieved the strongest performance on held-out test data. At the selected operating threshold of **0.695**, the model achieved **ROC-AUC = 0.956**, **Average Precision = 0.699**, **Precision = 0.891**, **Recall = 0.321**, and **F1 = 0.472**. In practical terms, this means the model can generate a **high-confidence target list** for premium or high-cost campaigns.

To answer the second question, I built a customer segmentation model using KMeans clustering. Although a **7-cluster** solution scored highest on an internal clustering metric, it split the mainstream employed population too finely to be useful for marketing. I therefore recommend using the **6-cluster technical solution**, translated into **5 client-facing personas** by combining two overlapping mainstream working-adult groups.

The main implications for the client are:

- **Use CatBoost to prioritize likely higher-income prospects.** The model is accurate enough to support targeted campaigns where contact cost matters.
- **Treat threshold selection as a business lever.** The current threshold prioritizes precision; it can be adjusted if the client later prefers broader reach.
- **Use personas to tailor offers and messaging.** The market is not one uniform population; life stage and economic structure differ substantially across segments.
- **Do not over-interpret purely technical clusters.** The best client solution is the one that is easiest to act on, not just the one with the best internal metric.

The recommended end-state is straightforward: use **CatBoost for targeting**, and use the **5-persona framework** to customize message, offer, and channel strategy.

1. Objective and Approach

The client requested two deliverables:

1. a model to distinguish individuals with income **above \$50K** from those at or below that level, and

2. a segmentation framework to group the population for marketing purposes.

These tasks serve different but complementary goals. The classification model supports **audience selection**, while the segmentation model supports **campaign design**.

The project was approached in three stages:

- understand the structure of the weighted census data,
 - develop and compare candidate classification models,
 - build and refine a segmentation that is useful for client decision-making.
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2. What the Data Suggests

The dataset contains roughly **199,523 records** with demographic, education, employment, and economic variables, plus a survey weight and an income label. A key business implication emerged immediately: the high-income population is a **minority group**, representing only about **6.2%** of the data. This means that a useful model must do more than achieve high overall accuracy; it must identify a relatively rare group efficiently.

Another important finding is that the population is highly heterogeneous. Differences in age, labor-force participation, education, and capital income suggest that a single marketing message would be inefficient. This supported the need for both a targeting model and a segmentation layer.

To keep the analysis business-defensible, the survey **weight** was used to reflect population importance in training and summaries, but it was **not treated as a customer attribute**. This avoids using sampling structure as if it were a real marketing feature.

3. Classification Results and Recommendation

Two models were developed for the income prediction task:

- **Logistic Regression** as an interpretable benchmark
- **CatBoost** as the final high-performance model

The final comparison is shown below.

Model	Test ROC-AUC	Test Avg. Precision	Test Precision	Test Recall	Test F1
Logistic Regression	0.946	0.613	0.792	0.301	0.436
CatBoost	0.956	0.699	0.891	0.321	0.472

CatBoost outperformed the benchmark on every key metric, and the gains are meaningful rather than cosmetic. Most importantly, it improved **Average Precision**, which matters because the client is trying to find a relatively rare high-income group. It also improved **precision** while slightly improving **recall**, meaning the model produces a cleaner target list without sacrificing reach.

At the current threshold of **0.695**, the model is tuned for **precision-focused targeting**. This is appropriate when the client wants to limit false positives, for example in premium outreach, high-value direct campaigns, or other contexts where each contact has meaningful cost.

Recommendation

The client should use **CatBoost** as the primary classification model for identifying likely >50K individuals. Logistic Regression should be retained only as a benchmark and interpretability reference.

Key client takeaway

The central business question is not whether the high-income audience can be identified; it can. The more important question is **how much reach the client wants to trade for confidence**. That decision can be managed through threshold selection.

4. Segmentation Results and Recommendation

The clustering analysis showed that the population can be organized into distinct economic and life-stage groups. However, the technically best clustering solution was not the best client-facing solution.

A **7-cluster** solution produced the strongest internal separation metric, but in practice it fragmented the mainstream employed population into multiple highly similar groups and created an extra small niche segment with limited marketing value. The **6-cluster** solution was therefore selected as the better foundation for business use.

For the client, the most useful way to present the result is as **5 personas**, not as raw technical clusters:

Persona 1: Dependents / children / non-earners

Very young population with effectively no workforce participation. This is relevant mainly in household-oriented marketing rather than direct income-based targeting.

Persona 2: Older low-work / retired households

Older individuals or households with limited labor-force attachment. This segment is better approached with life-stage, value, and stability-oriented messaging.

Persona 3: Mainstream working households

The broadest employed population. This should be treated as the core mass-market segment for general campaigns.

Persona 4: Younger working adults

A younger employed group with meaningful wage activity. This segment is suitable for messaging around convenience, mobility, lifestyle, and upward economic progression.

Persona 5: Affluent capital-gains niche

A very small but high-value group characterized by high education and unusually large capital gains. This should be treated as a specialty micro-segment rather than a mainstream target group.

Recommendation

The client should use the **5-persona framework** for campaign planning. This is more actionable than a purely technical 6- or 7-cluster output and better aligned with how marketing teams typically work.

Key client takeaway

The segmentation result is valuable not because it creates many clusters, but because it shows that **income prediction alone is not enough**. A high-probability prospect may still need a very different message depending on life stage and economic profile.

5. Business Recommendations

Based on the combined results, I recommend the following:

5.1 Use CatBoost for audience prioritization

Use the final CatBoost model to score and rank prospects by likelihood of belonging to the >50K group.

5.2 Adjust threshold by campaign type

The current threshold is appropriate for high-confidence targeting. If the client later wants broader reach, the threshold can be lowered deliberately.

5.3 Pair targeting with persona-based messaging

Use the classification score to decide **who enters the target pool**, and use the persona framework to decide **what message, offer, and channel to use**.

5.4 Treat the affluent niche separately

The capital-gains-heavy group is too small for mass marketing, but it may be strategically useful for premium or specialized campaigns.

6. Conclusion

This project delivered both requested outputs: a classification model for identifying likely higher-income individuals and a segmentation framework for organizing the market into actionable groups.

The final recommendations are:

- **Deploy CatBoost** as the final targeting model
- **Retain Logistic Regression** as an interpretable benchmark
- **Use the 5-persona segmentation framework** for marketing strategy
- Combine both outputs so that the client can answer both **who to target** and **how to target them**

This combined approach is more useful for real campaign planning than either model on its own.

Appendix

Appendix A. Data Exploration and Preprocessing Details

A1. Dataset structure

- Approximately **199,523 observations**
- Demographic, education, employment, and economic variables
- One survey **weight** column
- One binary income label (>50K vs <=50K)

A2. Main exploratory findings

- The positive class is rare: about **6.2%** of the data
- The dataset contains many categorical variables

- Some categorical levels are sparse or infrequent
- The survey weight should be used analytically, not as a normal predictive attribute
- Segmentation outputs need manual review because some technical clusters are overlapping or very small

A3. Preprocessing decisions

- The survey weight was excluded from the feature matrix
- The survey weight was used as **sample weight** for classification and for weighted summaries
- Logistic Regression used **one-hot encoding** with rare-category handling
- CatBoost used **native categorical feature handling**
- Data was split into **train / validation / test** subsets

A4. Why these preprocessing choices were made

These preprocessing choices were intended to preserve interpretability, avoid feature misuse, and support fair comparison between models.

Appendix B. Classification Modeling Details

B1. Training algorithms

- **Logistic Regression:** weighted, regularized linear classifier
- **CatBoost:** gradient-boosted decision tree model with native categorical handling

B2. Evaluation procedure

Models were compared using:

- ROC-AUC
- Average Precision
- Precision
- Recall
- F1
- Validation-based threshold selection

Thresholds were chosen to maximize **precision subject to recall ≥ 0.30** , reflecting a precision-oriented marketing use case.

B3. Logistic Regression results

Threshold: 0.585

Metric	Validation	Test
ROC-AUC	0.940	0.946

Average Precision	0.608	0.613
Precision	0.773	0.792
Recall	0.301	0.301
F1	0.434	0.436

B4. CatBoost results

Threshold: 0.695

Metric	Validation	Test
ROC-AUC	0.950	0.956
Average Precision	0.680	0.699
Precision	0.858	0.891
Recall	0.304	0.321
F1	0.449	0.472

B5. Why CatBoost was selected

CatBoost improved both ranking quality and final targeting quality. The strongest gain was in **Average Precision**, which is especially important for identifying a rare positive class.

Appendix C. Segmentation Modeling Details

C1. Clustering approach

KMeans clustering was evaluated across multiple values of k . Candidate solutions were reviewed using both internal separation metrics and client usefulness.

C2. Why $k = 7$ was not selected for business use

Although $k = 7$ achieved the highest silhouette score, it:

- split mainstream employed adults into multiple very similar groups,
- added an extra small niche cluster with limited practical value,
- produced a more fragmented result than the client needs.

C3. Why $k = 6$ was selected

The $k = 6$ solution was simpler and more interpretable. It still contained overlap among mainstream employed clusters, so the final client recommendation consolidates those into one broader persona.

C4. Technical-to-business translation

Technical clusters were translated into business personas so the segmentation could be used by the client's marketing team without requiring direct interpretation of raw clustering output.

Appendix D. Additional Interesting Findings

D1. Threshold choice is a business lever

Once model quality is strong, the threshold becomes a strategic choice rather than just a technical setting. Higher thresholds improve precision but reduce reach.

D2. Not every technical segment should be a business segment

The project showed that internal clustering metrics alone are not sufficient for client decision-making. Simplicity and actionability matter.

D3. The affluent niche is real but small

A very small, capital-gains-heavy segment appeared consistently across clustering solutions. It likely represents a real subpopulation, but it is too small to serve as a mass-market segment.

Appendix E. Limitations and Next Steps

E1. Limitations

- The data is census-derived and may not perfectly match the client's live prospect universe
- Segmentation is unsupervised, so segment value has not yet been validated against campaign response
- The current threshold reflects a precision-oriented objective; other campaign types may justify different cutoffs

E2. Suggested next steps

- test score bands instead of relying on one hard threshold
- validate persona usefulness with campaign response or A/B testing
- enrich the analysis with client-specific behavioral or transaction data if available
- calibrate thresholds by campaign economics

Appendix F. References

- Project-provided census data file and column description file
- scikit-learn documentation for preprocessing, Logistic Regression, model evaluation, and KMeans clustering
- CatBoost documentation for categorical feature handling and classifier training
- Standard references for ROC-AUC and precision-recall evaluation in imbalanced classification